GRAPH OF THOUGHTS SIGNAL MODELING FOR SEQUENTIAL RECOMMENDATION

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ABSTRACT

Large language models (LLMs) have recently attracted significant attention in sequential recommendation due to their strong reasoning and open-domain knowledge. However, most existing approaches rely on simple input–output prompting, which limits their ability to capture fine-grained user preferences and task-specific information. To address these limitations, we propose GOT4Rec, a novel framework that integrates the Graph of Thoughts (GoT) reasoning strategy into sequential recommendation. Specifically, GOT4Rec decomposes user sequences into short-term, long-term, and collaborative preference signal components, and models them with LLM-guided generation and aggregation operations. This preference signal-inspired multi-path reasoning enables fine-grained preference modeling and accurate reconstruction of next-item signals. Extensive experiments on multiple real-world datasets demonstrate that GOT4Rec outperforms strong baselines by an average of 32.21%.

Index Terms— Recommender Systems, Graph of Thoughts, Large Language Models

1. INTRODUCTION

Sequential recommendation has long been a significant research field, with numerous methods proposed to explore temporal patterns and behavioral signals within user sequences [1, 2, 3]. Despite advancements, performance remains limited by the closed and sparse nature of standard datasets. Recently, large language models (LLMs) have attracted attention for their expansive knowledge and reasoning abilities, inspiring research integrating LLMs into sequential recommendation [4, 5], aiming to capture user interests from interaction signals while leveraging LLMs' comprehensive real-world knowledge for recommendations.

Although LLM-based methods have shown improvements, most rely on simple input–output prompting [7, 8], leaving their reasoning largely untapped and producing task-irrelevant results. To address this, advanced reasoning strategies have been proposed. One representative example is SLIM [6], a knowledge distillation model which employs chain-of-thought (CoT) [9] prompting to enable step-by-step reasoning in sequential recommendation, transferring knowledge from a teacher model to a student model. However, SLIM's reasoning remains shallow and cannot fully model complex preference signal structures, failing to disentangle short-term/long-term preferences or spatio-temporal patterns. As illustrated in Figure 1, SLIM's CoT-based reasoning yields a recommendation focused solely on snack bars, whereas the user's actual preference centers on a fruit nut mix, a category better captured through short-term preference modeling. This example highlights SLIM's limitations in handling diverse and multifaceted preference structures.

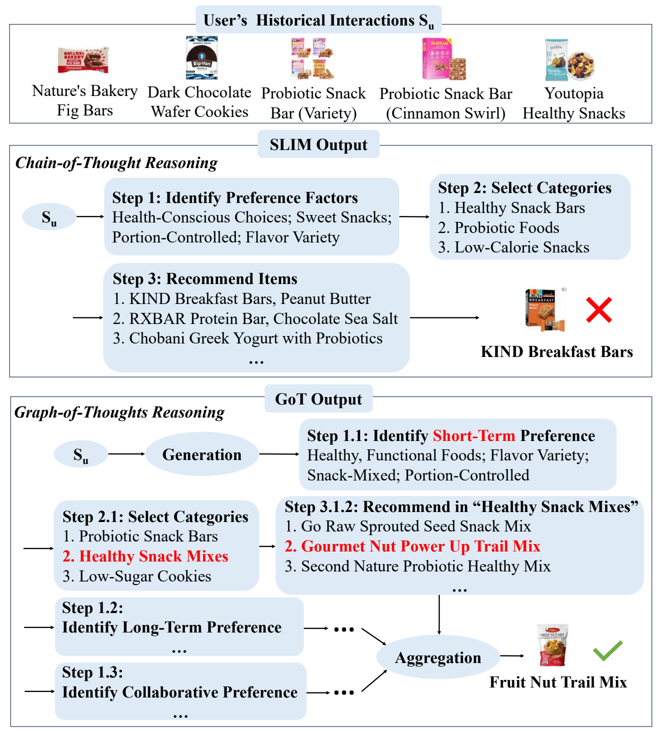


Fig. 1. Comparison of the LLM output and recommendations generated by SLIM [6] and GOT4Rec.

Therefore, the core challenge lies in **effectively integrating diverse user preferences into a cohesive reasoning framework**, turning sequential recommendation into a multi-faceted task requiring advanced reasoning capabilities. Which leads us to the graph of thoughts (GoT) approach, a method that offers a more structured and effective solution by decomposing the problem into more manageable components. Unlike CoT, which employs one single chain of thought, GoT models the reasoning process as a graph, allowing it to break down the complex user preference reasoning tasks into smaller, more tractable sub-tasks to solve them individually, and then integrate the results to form a comprehensive solution. As illustrated in Figure 1, GoT enables the LLM to explore multiple relevant product categories, such as Probiotic Snack Bars, Healthy Snack Mixes and Low-Sugar, Low-Carb Cookies before synthesizing a final recommendation. This multi-path reasoning ultimately leads to the correct prediction of the ground truth item: fruit nut mix. In contrast, SLIM's CoT-based reasoning remains narrowly focused on snack bars, highlighting the limitations of linear reasoning in capturing diverse user intents.

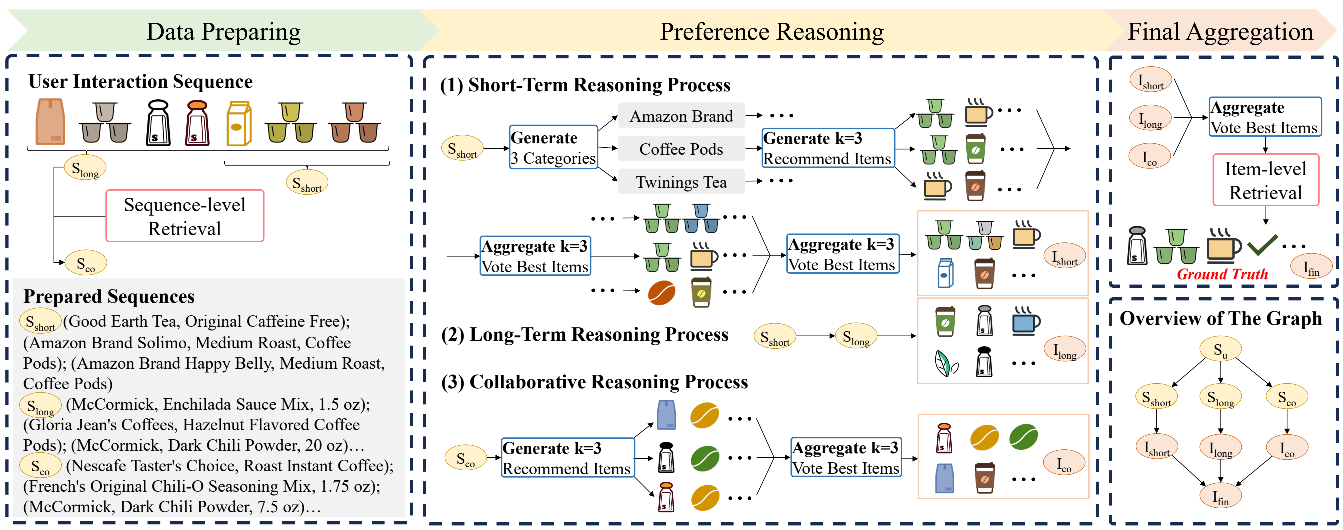


Fig. 2. An example of GOT4Rec. The user interaction sequence is split for different reasoning types: recent items for short-term preferences, the full sequence for long-term preferences, and similar users for collaborative reasoning. These thoughts are generated in a graph and aggregated to integrate multiple preference signals.

To address these challenges, we propose GOT4Rec, a novel sequential recommendation method based on the Graph of Thoughts (GoT) framework. GOT4Rec leverages LLM reasoning while integrating short-term, long-term, and collaborative preference signals from user sequences. Unlike prior methods that treat the sequence as a monolithic input, GOT4Rec disentangles these dimensions and applies multi-path LLM reasoning to produce more accurate and interpretable recommendations. Our main contributions are as follows:

- To our knowledge, this is among the first works to formulate sequential recommendation under the Graph of Thoughts reasoning framework.
- We propose GOT4Rec, a GoT-based approach that exploits the structured reasoning capability of LLMs to model user preferences across temporal and collaborative signals.
- Experiments on public datasets show that GOT4Rec consistently outperforms traditional neural models, recent LLM-based baselines, and alternative LLM reasoning strategies.

2. METHODOLOGY

In this section, we propose GOT4Rec, a sequential recommendation method that leverages the reasoning abilities of LLMs under the Graph of Thoughts framework. Given a user's history $S_u = \{i_1, \ldots, i_{n-1}\}$, the goal is to predict the next item i_n. Each item contains title and description information.

2.1. Overview of Thought Graph

We formalize GOT4Rec as a tuple (G, T), where $G = (V, E)$ is a directed graph representing the reasoning process and T denotes thought transformations. Each vertex $v \in V$ corresponds to a thought at a recommendation step, and an edge $(v_1, v_2) \in E$ indicates that v_2 is generated from v_1. We adopt the generation and aggregation transformations from GoT:

$$T_G(v) \to \{v_1^+, \ldots, v_k^+\}, \quad E_+ = \{(v, v_i^+)\}_{i=1}^k, \tag{1}$$

$$T_A(\{v_1, \ldots, v_k\}) \to v^+, \quad E_+ = \{(v_i, v^+)\}_{i=1}^k, \tag{2}$$

where T_G expands a thought into multiple candidates and T_A aggregates multiple thoughts into a new vertex v^+.

Figure 2 illustrates the graph decomposition of GOT4Rec. Given a user interaction sequence S_u, we derive short-term, long-term, and collaborative preference signals. Based on these signals, the LLM generates top-N candidate items, which are then aggregated and voted to produce the final recommendation.

2.2. Short-Term & Long-Term Reasoning Process

Understanding both short-term dynamics and long-term user preferences is essential for effective recommendation [10, 11]. In GOT4Rec, we guide the LLM to separately model these two preference signal components using zero-shot CoT prompts. For short-term reasoning, we select the most recent interactions from the user sequence S_u as S_{short}. We then apply a two-step generation strategy within the transformation T_G. It involves two key steps:

- Step 1. Summarize user's short-term preference signals based on the given S_{short}.
- Step 2. Based on the preferences signals in step 1, summarize three categories of products the user is inclined to prefer.

For each category, the LLM generates N candidate items. This process can be repeated k times to increase diversity. The resulting k item sets are aggregated via LLM-based voting (transformation T_A) to produce one representative set per category. A final round of voting across the three categories yields the top-N short-term preference items I_{short}.

For long-term reasoning, we input the entire sequence S_u and follow a similar two-step CoT-based prompting strategy, allowing the model to abstract stable user preference signals. The final long-term item set I_{long} is generated through aggregation T_A.

2.3. Collaborative Reasoning Process

To incorporate collaborative signals, we adopt a retrieval-augmented strategy. The target user's sequence is encoded using the all-mpnet-base-v2 [12] model to form a dense vector, which is used to retrieve a set of similar user sequences S_{co}. Details of the retrieval process are discussed in the following section. We then prompt the LLM using the following two-step zero-short CoT strategy:

Dataset	Metric	GRU4Rec	SASRec	BERT4Rec	PALR	TALLRec	SLIM	CoT4Rec	CoT	CoT-SC	ToT	GOT4Rec	Improv.
Games	H@5	0.0390	0.0776	0.0839	0.0630	0.0660	0.0674	0.0692	0.0644	0.0653	0.0489	**0.0894**	6.56%
	H@10	0.0545	0.0953	0.1009	0.0893	0.0967	0.0981	0.0991	0.0988	0.1013	0.0712	**0.1167**	15.20%
	H@20	0.0765	0.1227	0.1302	0.1157	0.1233	0.1302	0.1346	0.1347	0.1253	0.1087	**0.1361**	1.04%
	N@5	0.0266	0.0528	0.0563	0.0385	0.0401	0.0392	0.0403	0.0408	0.0421	0.0304	**0.0621**	10.30%
	N@10	0.0317	0.0586	0.0617	0.0423	0.0497	0.0517	0.0528	0.0519	0.0537	0.0377	**0.0710**	15.07%
	N@20	0.0372	0.0655	0.0681	0.0475	0.0562	0.0588	0.0601	0.0609	0.0622	0.0471	**0.0760**	11.60%
Food	H@5	0.0213	0.0335	0.0321	0.0337	0.0390	0.0382	0.0400	0.0396	0.0443	0.0273	**0.0742**	67.49%
	H@10	0.0329	0.0407	0.0452	0.0487	0.0513	0.0593	0.0613	0.0581	0.0597	0.0390	**0.0972**	58.56%
	H@20	0.0525	0.0549	0.0540	0.0657	0.0747	0.0741	0.0760	0.0748	0.0753	0.0533	**0.1090**	43.42%
	N@5	0.0133	0.0286	0.0271	0.0216	0.0243	0.0247	0.0257	0.0253	0.0276	0.0182	**0.0492**	72.03%
	N@10	0.0170	0.0309	0.0311	0.0235	0.0284	0.0290	0.0298	0.0311	0.0326	0.0219	**0.0567**	73.93%
	N@20	0.0219	0.0337	0.0324	0.0267	0.0318	0.0335	0.0358	0.0352	0.0366	0.0255	**0.0597**	63.11%
Home	H@5	0.0095	0.0132	0.0139	0.0130	0.0123	0.0130	0.0133	0.0118	0.0133	0.0047	**0.0192**	38.13%
	H@10	0.0164	0.0179	0.0204	0.0167	0.0183	0.0189	0.0194	0.0177	0.0223	0.0100	**0.0299**	34.08%
	H@20	0.0274	0.0262	0.0299	0.0213	0.0233	0.0241	0.0249	0.0230	0.0270	0.0147	**0.0337**	12.71%
	N@5	0.0058	0.0097	0.0108	0.0080	0.0081	0.0077	0.0081	0.0073	0.0080	0.0031	**0.0122**	12.96%
	N@10	0.0080	0.0105	0.0120	0.0094	0.0096	0.0098	0.0101	0.0093	0.0109	0.0049	**0.0157**	30.83%
	N@20	0.0108	0.0134	0.0148	0.0101	0.0113	0.0106	0.0109	0.0106	0.0121	0.0060	**0.0167**	12.84%

Table 1. Recommendation performance. The best performance is highlighted in **bold** and the runner-up is highlighted by underlines. Improv. indicates relative improvements over the best baseline in percentage.

- Step 1. Summarize the shared collaborative preference signals between the current user and other users by S_{co}.
- Step 2. Based on the summarized signals, recommend N items selected from S_{co} that the current user is likely to prefer.

This is repeated three times to obtain multiple sets, which are aggregated via T_A to form the collaborative recommendation set I_{co}.

2.4. Final Alignment and Integration

The three sets are first fused via the aggregation transformation T_A, which prompts the LLM to consolidate the information and produce the final top-N recommendation set I_{fin}. Subsequently, to ensure alignment with the evaluation dataset, an item-level retrieval step is conducted, matching each item in I_{fin} to its closest real counterpart. The same voting prompt is employed during this integration process.

2.5. Two-Level Retrieval Module

To support retrieval-augmented generation, we design a two-level retrieval module providing LLMs with context for collaborative reasoning and dataset consistency. At the sequence level, we retrieve top-k similar user sequences from a pre-encoded vector base $\mathcal{V}$seq using embeddings f_{emb}:

$$\mathcal{S}^k_{\mathcal{V}_{\text{seq}}}(i_t) \triangleq \arg \max^k_{\boldsymbol{v}\in\mathcal{V}_{\text{seq}}} \text{sim}(\frac{1}{m}\sum_{t=1}^{m} f_{\text{emb}}(i_t), \boldsymbol{v}), \tag{3}$$

which are injected into the LLM prompt for collaborative preference reasoning. At the item level, to align free-form outputs i_{q} with canonical dataset items, we retrieve top-k matches from $\mathcal{V}$item above a similarity threshold τ:

$$\mathcal{I}^k_{\mathcal{V}_{\text{item}}}(i_{\text{q}}) \triangleq \arg \max^k_{\substack{\boldsymbol{v}\in\mathcal{V}_{\text{item}} \\ \text{sim}(f_{\text{emb}}(i_{\text{q}}),\boldsymbol{v})\geq\tau}} \text{sim}(f_{\text{emb}}(i_{\text{q}}), \boldsymbol{v}), \tag{4}$$

ensuring generated recommendations are both contextually grounded and semantically consistent.

2.6. Inference Latency and Volume

We compare inference latency and volume across reasoning frameworks. Latency is the number of hops to reach the final thought, and volume is the number of intermediate thoughts contributing to it. We assume each thought is generated in $O(1)$ time. In CoT, reasoning is sequential, so both latency and volume scale as N. CoT-SC uses k parallel chains, reducing both to N/k. For a k-ary ToT tree, both latency and volume are $\log_k N$. GoT combines a k-ary tree with its inverse, resulting in latency $\log_k N$ but volume N, since all intermediate thoughts are aggregated. GOT4Rec enables multiple parallel reasoning paths; its graph structure keeps latency manageable. In practice, batching and prompt caching further improve scalability.

3. EXPERIMENTS

3.1. Experiment Settings

Datasets. We evaluate on three Amazon Reviews'23 categories [13]: Video Games (**Games**), Grocery and Gourmet Food (**Food**), and Home and Kitchen (**Home**). Reviews are treated as timestamp-ordered user-item interactions. We retain users with 6–20 interactions and items with at least 5 interactions, and randomly sample 3,000 users from each dataset three times [6]. Following the leave-one-out protocol, the last interaction is used for testing, the second-last for validation, and the remainder for training; both training and validation segments are used during testing.

Baselines. To demonstrate the effectiveness of our proposed method, we select two groups of recommendation baselines. The first group comprises both neural and LLM-based sequential recommenders, including GRU4Rec [14], SASRec [1], BERT4Rec [15], PALR [8] and TALLRec [16]. The second group contains models that utilize LLM reasoning strategies, including SLIM [6], CoT4Rec [17], Chain-of-Thought (CoT) [9], Multiple CoTs (CoT-SC) [18] and Tree of Thoughts (ToT) [19].

Implementation Details. We use Llama3-8B-Instruct [20] as the backbone. GRU4Rec, SASRec, BERT4Rec, TALLRec, and CoT4Rec follow official implementations, while PALR is self-implemented and LLM reasoning is integrated into the GoT framework. For SLIM, we replace ChatGPT with Llama3-70B-Instruct to reduce API costs. For all methods, up to 10 similar items are retrieved using Faiss [21] with threshold $\tau = 0.75$. Hyperparameters are tuned for optimal performance.

Evaluation Metrics. We evaluate with hit rate (HR) and normalized discounted cumulative gain (NDCG), reporting HR@K and NDCG@K for $K \in \{5, 10, 20\}$. Each recommendation is compared to all items. Results are averaged over three runs.

3.2. Results

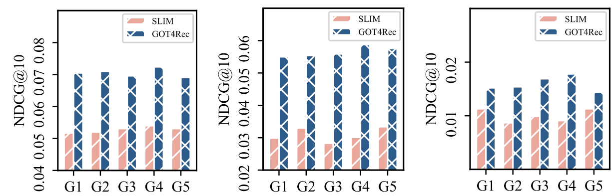
Overall Performance. Table 1 compares our GOT4Rec method with various neural sequential models and LLM reasoning strategies, yielding several key observations. Traditional neural and LLM-based models generally achieve moderate performance: BERT4Rec remains competitive due to its bidirectional self-attention capturing sequential transitions, but its limited semantic understanding and reasoning restrict modeling abstract user intent, while PALR and TALLRec underperform because of shallow prompting and lack of multi-step inference. Among LLM reasoning strategies-based methods, CoT-SC performs best outside our model, benefiting from multi-path reasoning and self-consistency, whereas SLIM, CoT4Rec, and CoT-SC struggle due to shallow reasoning and limited preference modeling, and ToT's hierarchical structure misaligns with sequential user interactions. In contrast, GOT4Rec achieves state-of-the-art performance across all datasets, e.g., on Food, it improves NDCG@10 by 73.93% and HR@5 by 67.49% over CoT-SC, effectively modeling short-term preference signals and brand-category affinities while using fewer parameters. By integrating short-term, long-term, and collaborative preferences signals through structured reasoning, GOT4Rec consistently produces more accurate, diverse, and interpretable recommendations than both traditional neural and recent LLM-based models.

Dataset	Metric	w/o Short.	w/o Long.	w/o Co.
Games	H@5	$0.0819_{\downarrow 8.39}$	$0.0871_{\downarrow 2.57}$	$0.0774_{\downarrow \underline{13.42}}$
	H@10	$0.1083_{\downarrow 7.20}$	$0.1121_{\downarrow 3.94}$	$0.1015_{\downarrow \underline{13.02}}$
	H@20	$0.1262_{\downarrow 7.27}$	$0.1337_{\downarrow 1.76}$	$0.1170_{\downarrow \underline{14.03}}$
	N@5	$0.0512_{\downarrow 17.55}$	$0.0573_{\downarrow 7.73}$	$0.0527_{\downarrow \underline{15.14}}$
	N@10	$0.0597_{\downarrow \underline{15.92}}$	$0.0654_{\downarrow 7.89}$	$0.0606_{\downarrow 14.65}$
	N@20	$0.0642_{\downarrow 13.53}$	$0.0709_{\downarrow 6.71}$	$0.0645_{\downarrow \underline{15.13}}$
Food	H@5	$0.0591_{\downarrow \underline{20.35}}$	$0.0867_{\uparrow 16.85}$	$0.0687_{\downarrow 7.41}$
	H@10	$0.0867_{\downarrow \underline{10.80}}$	$0.0933_{\downarrow 4.01}$	$0.0880_{\downarrow 9.47}$
	H@20	$0.1003_{\downarrow \underline{7.98}}$	$0.1063_{\downarrow 2.48}$	$0.1013_{\downarrow 7.06}$
	N@5	$0.0362_{\downarrow \underline{26.42}}$	$0.0426_{\downarrow 13.41}$	$0.0437_{\downarrow 11.18}$
	N@10	$0.0453_{\downarrow \underline{20.11}}$	$0.0509_{\downarrow 10.23}$	$0.0500_{\downarrow 11.82}$
	N@20	$0.0487_{\downarrow \underline{18.43}}$	$0.0543_{\downarrow 9.05}$	$0.0534_{\downarrow 10.55}$
Home	H@5	$0.0179_{\downarrow 6.77}$	$0.0166_{\downarrow \underline{13.54}}$	$0.0190_{\downarrow 1.04}$
	H@10	$0.0284_{\downarrow 5.02}$	$0.0292_{\downarrow 2.34}$	$0.0270_{\downarrow \underline{9.70}}$
	H@20	$0.0322_{\downarrow 4.45}$	$0.0318_{\downarrow 5.64}$	$0.0309_{\downarrow \underline{8.31}}$
	N@5	$0.0102_{\downarrow \underline{16.39}}$	$0.0103_{\downarrow 15.57}$	$0.0114_{\downarrow 6.56}$
	N@10	$0.0136_{\downarrow \underline{13.38}}$	$0.0144_{\downarrow 8.28}$	$0.0140_{\downarrow 10.83}$
	N@20	$0.0146_{\downarrow \underline{12.57}}$	$0.0150_{\downarrow 10.18}$	$0.0149_{\downarrow 10.78}$

Table 2. Ablation study, the performance difference (%) with GOT4Rec is also provided. "Short.", "Long.", and "Co." denote short-term, long-term and collaborative steps, respectively.

Ablation Study. We analyze the impact of GOT4Rec components in Table 2. The full model consistently outperforms all ablated variants, confirming the effectiveness of integrating short-term, long-term, and collaborative preference signals. The study also reveals that the importance of each signal component varies by dataset. In Games, collaborative signals are most impactful, likely due to users' reliance on peer behavior for brand or category decisions. In Food, short-term signals dominate, reflecting users' immediate needs. This aligns with our inference that users' recent preferences heavily influence their choices in food products. In Home, long-term and collaborative signals matter more, while short-term influence is minimal, suggesting more deliberate and stable decision-making patterns.

Data Sparsity Analysis. We analyze the impact of data sparsity by grouping users according to interaction density. Users are ranked by interaction frequency and divided into five equal-sized groups from sparse to dense. We evaluate GOT4Rec and SLIM on each group. As shown in Figure 3, GOT4Rec consistently outperforms SLIM across all groups. These results highlight the robustness of our signal-based approach and its ability to maintain high performance even under limited interaction signals, effectively addressing this key challenge in sequential recommendation.



(a) Games (b) Food (c) Home

Fig. 3. Analysis for different sparse-level users. The sparsity decreases from user group G1 to user group G5.

Popularity Bias Analysis. To assess recommendation novelty, we calculate EFD@10 and EPC@10 [22]. EFD measures the expected inverse frequency of seen relevant items, while EPC captures the number of relevant recommendations that were previously unknown to the user. Together, these metrics reflect both novelty and diversity, helping assess a model's ability to balance popular and long-tail content. As in Table 3, GOT4Rec consistently outperforms CoT, SLIM and CoT4Rec on both metrics. From a signal-processing perspective, this suggests GOT4Rec is more effective in capturing long-tail preference signals beyond dominant popularity trends.

Dataset	Metric	CoT	SLIM	CoT4Rec	GOT4Rec
Games	EFD@10	4.8919	4.9372	5.0218	**5.0929**
	EPC@10	0.3885	0.3916	0.3947	**0.3998**
Food	EFD@10	6.4721	6.3997	6.5443	**6.8671**
	EPC@10	0.4750	0.4613	0.4870	**0.5035**
Home	EFD@10	4.1911	4.3010	4.3619	**4.6019**
	EPC@10	0.3110	0.3216	0.3305	**0.3450**

Table 3. Popularity bias metrics, the best score is in **bold**.

4. CONCLUSION

In this paper, we propose GOT4Rec, a sequential recommendation method that optimally leverages the reasoning capabilities of LLMs to extract and integrate short-term, long-term, and collaborative preference signals under the graph of thoughts (GoT) framework. Experiments on real-world datasets demonstrate that GOT4Rec outperforms existing neural sequential models and LLM reasoning strategies. Further analysis reveals that GOT4Rec effectively integrates multiple signal components contained within user sequences, leading to superior recommendations.

5. ACKNOWLEDGEMENTS

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